



JAIPUR NATIONAL UNIVERSITY

Semester/Year End/Back Examination Dec.-2024

Name of School:	School of Engineering & Technology
Name of Programme / Course:	B.Tech. (Computer Science)
Semester/Year:	III Semester
Paper Code:	SCSSAGE301T22
Title of Paper:	Basic Computer Networks - Everyone Must Know
Time: 3 Hours	Max. Marks: 60

INSTRUCTIONS / NOTES FOR THE EXAMINEE

1. This question paper consists of two sections. Section A and Section B.
2. Section A consists of seven questions and you have to attempt any five questions.
3. Section B consists of six questions and you have to attempt any four questions.
4. Marks allotted are mentioned against each question.
5. Draw neat and clean diagrams wherever required.

Question Number	SECTION-A	Marks Allotted
Q.1	Describe the primary components of Computer Networks.	
Q.2	Describe the types of line configurations in Network Communication.	
Q.3	Provide an overview of the TCP/IP Reference Model, listing its layers and their functions.	
Q.4	Explain the HTTP Protocol and its role in Web Communication.	
Q.5	Explain the role of SMTP in Email Communication.	
Q.6	Describe the function of Telnet and its common use cases.	
Q.7	Explain the concept of Network Architecture and its significance.	
		5X4=20

SECTION-B

~~Q.1~~ Define Network Protocols and explain the role of protocols in enabling communication between devices in a network.

~~Q.2~~ Compare different Network Topologies (Bus, Star, Ring, and Mesh) in terms of structure, cost, and suitability for different environments.

~~Q.3~~ Describe the OSI reference model in detail, focusing on the purpose and functionality of each layer.

4X10=40

~~Q.4~~ Explain the components of Data Communication, detailing how each component contributes to successful data transfer.

Q.5 Describe the Domain Name System (DNS) in detail, including how it translates domain names into IP addresses and its importance in internet functionality.

Q.6 Describe the concept of the Internet, its basic architecture, and how it enables global communication. Discuss some of the key services provided over the Internet.